FINAL EVALUATION

PART 2: Analysis of RNA-seq or single-cell data-set | BESE-300

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1 Description of Data: Curation and Exploration

The present report analyzes the series GSE75417, part of the STATegra dataset (Gomez-Cabrero et al. 2019). The dataset consists of the gene expression (based on RNA-seq) of B3 cell differentiation in mice, divided into two groups: Control (Ctr) and treated with Tamoxifen for Ikaros induction (Ik) across 6 time points (0, 2, 6, 12, 18, and 24 hours).

The raw file consists of 38 columns and 12762 rows, where each row represents a gene and each column a sample. The first column contains the ID, the second the MGI (Mouse Genome Informatics) gene symbol (261 not available), while the rest of the columns contain the expression values for each sample (Table 1).

Table 1: Data-Set (first 3 rows and columns)

X	MGI_Symbol	Batch_1_Ctr_0H
ENSMUSG00000000001	Gnai3	13.594162
ENSMUSG00000000028	Cdc45	13.812612
ENSMUSG00000000031	H19	5.607099

As seen in Figure 1, there are no negative values, the minimum value is approximately 1, the maximum is approximately 22 (log-transformed), and every column possesses a similar median (around 11) and spread (normalized). Terms inside the filename give us more information about the preprocessing, like CQN (Conditional Quantile Normalization) and ComBat (to remove batch effects). For this particular case, after checking the documentation and taking into account the data origin, we can assume that the data was properly cleaned of low-expressed genes, abnormal samples/errors, and batch effects, and that it has been log-transformed and normalized.

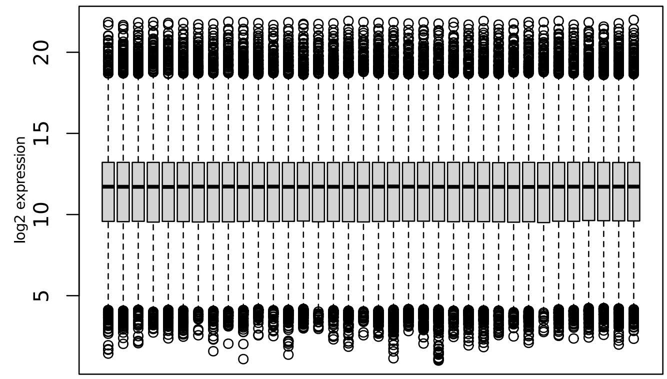

Figure 1: Normalized expression, 36 samples

1.1 Quality Control: Experimental Design and Principal Component Analysis

Among the columns, every sample name encodes the batch, condition, and time point (e.g., Batch_1_Ctr_0H). The design is unbalanced: batch 2 is only present at time 0H, while batch 3 is missing at 0H but present for the other time points. However, every condition maintains 3 replicates (Table 2).

Table 2: Experiment design

Batch	0H	2H	6H	12H	18H	24H
1	X	X	X	X	X	X
2	X					
3		X	X	X	X	X
4	X	X	X	X	X	X

As we have many variables, a PCA of all 36 samples—colored by time and shaped by condition—was created to visually check the formation of groups. As expected, in Figure 2 we note that all the control dots are grouped together at the bottom left corner, mainly varying along the second principal component (10.6%), indicating that there are no major transcriptomic differences among the control samples over time (uninduced B3 cells should not differentiate). When analyzing the treatment group (IK), it is easy to divide most of the time points into subgroups by their distance along the first principal component (46.8%). This means that for the IK group, as time progressed and the cells matured, their gene expression profiles became increasingly distinct. This confirms that we will be able to distinguish the biomarkers involved in the differentiation of

B-cells. Furthermore, because the samples group by biological conditions rather than batches, we can assume that the preprocessing of the dataset using ComBat was successful and the data is of high quality.

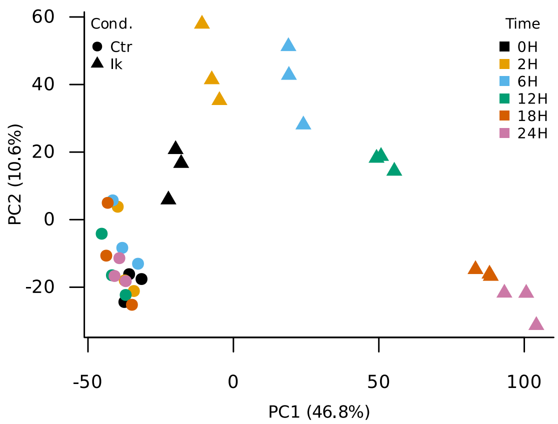

Figure 2: PCA of all 36 samples, colored by time, shaped by condition

2 Description of Statistics Used to Validate Associations

Because the dataset consists of three replicates per condition and the expression values were preprocessed into a log-scale matrix (via CQN and ComBat) rather than raw counts, traditional count-based differential expression tools like DESeq2 or NOISeq were not appropriate. Instead, limma (Ritchie et al. 2015) was utilized to fit a linear model to the expression data of each gene. To compare the 24H and 0H time points directly, a specific contrast was applied to the model. Statistical significance was calculated using limma's Empirical Bayes method, useful in this case due to the small sample sizes.

2.1 Differential Expression Model

The top biomarkers were sorted by adjusted p-value in order to filter out background noise and focus mainly on the genes we are most certain changed. The resulting analysis yielded the top 10 most statistically significant differentiation markers, as seen in Table 3. The log2FC (Log2 Fold Change) column indicates the magnitude and direction of the change; a positive value represents upregulation at 24H, while a negative value represents downregulation. Because the scale is logarithmic (base 2), a log2FC of 3 translates to an 8-fold change in actual expression. Notably, nine of the top ten markers are heavily upregulated, indicating a massive activation of genetic networks during B-cell maturation. Genes such as Dok3 (Downstream of kinase 3) and Tifa (TRAF-interacting protein with a FHA domain) display log2FC values of 3.67 and 4.61, meaning their expression levels increased approximately 13 and 24-fold, respectively. This is expected, as there are reports of these genes being involved in B-cell signaling and differentiation (Loh et al. 2020; Fu et al. 2026).

Table 3: Top differentiation markers (24H vs 0H)

Gene	log2FC(24H)	t	adj.P
D930015E06Rik	3.48	39.01	1.5e-20
Dok3	3.67	36.96	3.1e-20
Cerk	4.93	35.52	5.7e-20
Agpat9	3.55	34.79	7.2e-20
Tifa	4.61	34.36	7.9e-20
Otub2	4.97	30.12	1.9e-18
Ass1	−3.44	−29.54	2.6e-18
Tnfrsf26	3.50	29.39	2.6e-18
Syk	2.50	29.12	2.8e-18
Neil1	4.99	29.07	2.8e-18

To corroborate these statistical results, a volcano plot was created to visualize the differential expression of all genes between 24H and 0H (Figure 3). The x-axis represents the log2 fold change, while the y-axis represents the -log10 of the adjusted p-value. We can see that nine of the top ten markers are located in the extreme upper-right quadrant, visualizing them as the most statistically significant upregulated genes. Conversely, Ass1 is clearly visible in the upper-left quadrant as a highly significant downregulated marker. Furthermore, the heatmap of these top 10 markers (Figure 4) demonstrates perfect sample clustering. The dendrogram groups the upregulated genes tightly with the 24H IK samples, while the single downregulated gene (Ass1) shows high expression exclusively in the control and 0H IK samples.

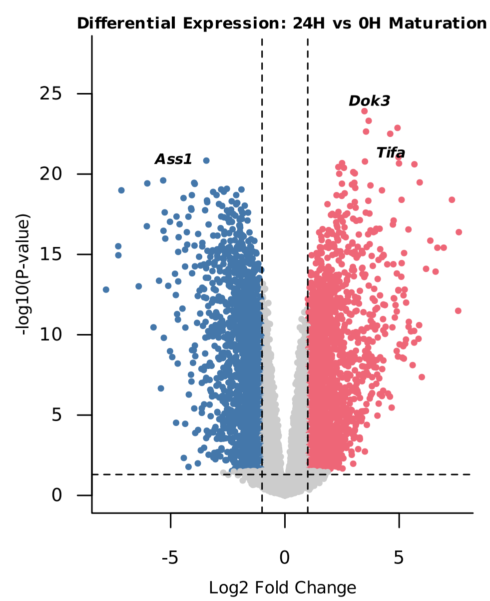

Figure 3: Differential Expression: 24H vs 0H Maturation

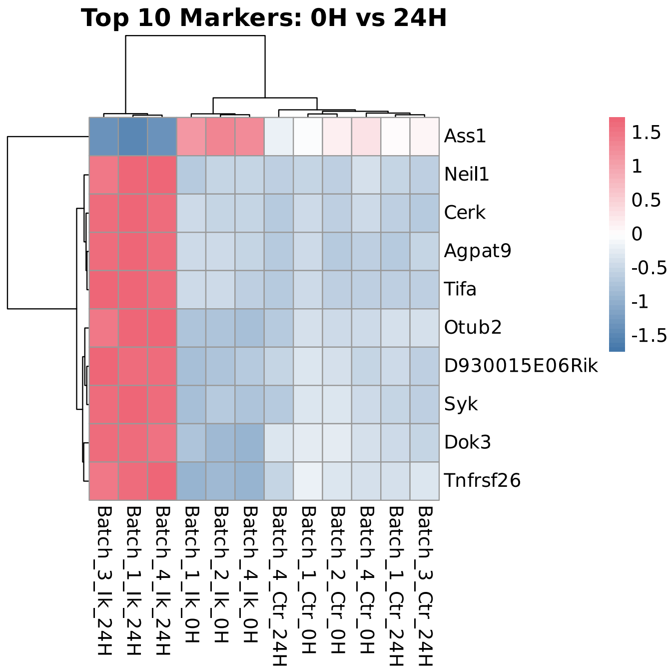

Figure 4: Top 10 markers: 0H vs 24H

2.2 Positive Control

Markers like Rag1 and Igll1 are well-known and highly useful for biologically validating the expression of other markers in B cells (Reynaud et al. 2008). As seen in Figure 5, Igll1 (which is expressed predominantly in large pre-B cells) maintains its expression level throughout the entire experiment in the control group, but drops significantly in the treated cells. Conversely, Rag1 is heavily upregulated in the treated group, which is the expected behavior for transitioning small pre-B cells. The correct behavior of these positive controls confirms the biological validity of the experiment and the reliability of our novel biomarker discoveries. While the existence of alternative analytical approaches, such as a Difference-in-Differences (DiD) F-test, is acknowledged, the current approximation successfully highlights critical markers driving the differentiation process of B3 cells under tamoxifen induction.

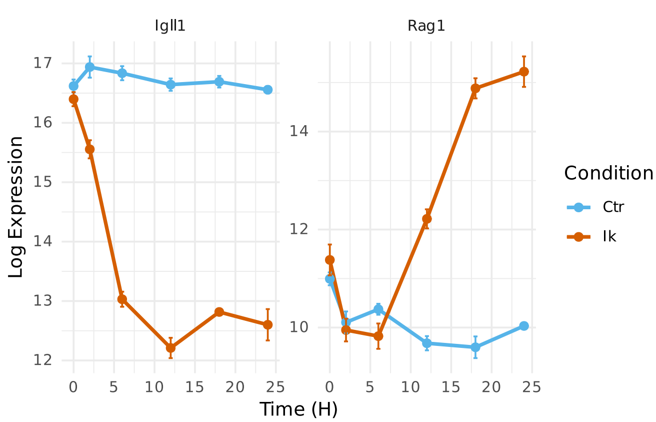

Figure 5: Expression Dynamics of Canonical Maturation Markers

References

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